

Name: _____ Student ID: _____

1. (10 points) Expand the following logarithmic functions as much as possible:

(a) $\ln \left(\frac{x^3 \sqrt{y+1}}{(2y)^2} \right)$

(b) $\ln \left(\frac{(x^2 + 3y)^4}{e^{2x} y^3} \right)$

Bonus (+1 point): Take the derivative of (a) w.r.t. $\ln(x)$

2. (10 points) Kelsey has utility function $U(x, y) = x^{1/4}y^{3/4}$, and she faces prices $p_x = 1$ and $p_y = 2$. How much income does she need to achieve utility level $U = 8$?

(Helpful calculations: $2^{3/4} \approx 1.68$; $3^{3/4} \approx 2.28$. But you may leave your answer un-simplified)

Bonus: Who is the current Chair of the UCLA Economics Department?